Doc. 308 – Λ as a Reading Artefact: Ontological Status of the Dark Sector under Cosmological Degeneracy

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Preface and Context

This document has two cores that belong together.

Core 1 — The Two-Halves Synthesis (Stage B): Einstein 1911 calculated the deflection of light at the Sun from time dilation alone: $\alpha = 0.875''$ — half the measured value. Einstein 1915 (GR) yielded $\alpha = 1.750''$, because space *and* time curvature each contribute one half.

In FFGFT, $\tilde{T}\cdot m = 1$ (time curvature) and fractal path elongation (Doc. 182, space curvature) are *not* alternatives but **complementary halves of the same deflection** — each exactly $0.875''$, together exactly $1.750''$. This is not a fit but a consequence of K3 area invariance ($\gamma_{\text{PPN}} = 1$ exactly, Cassini passed). FFGFT *explains* why Einstein 1911 was missing half: he had only the time rate, not the space half.

Core 2 — Ontological Status of the Dark Sector: If redshift admits two readings — expansion ($a(t)$) or static geometric path stretching ($1 + z = e^{\xi x}$, Doc. 280) — and the classical data (SNe Ia, BAO, CMB) are degenerate between both (Doc. 267), then quantities that exist *only within one reading* have the status of bookkeeping quantities without reading-independent reference. Dark energy satisfies all three artefact conditions. Dark matter does not — it has reading-independent evidence (rotation curves, Bullet Cluster) and is an open geometric problem for FFGFT.

Connection of both cores: The space half of the Einstein deflection ($\tilde{T} \cdot m = 1$ + K3 area) is structurally the same as the potential that flattens rotation curves at accelerations $a < a_0 = cH_0/(2\pi)$ — a_0 comes from the same ξ^{10} chain (Stage A). Dark matter and dark energy are therefore not equally open: DM has geometric candidates in the corpus; DE has no reading-independent existence.

This document is not part of the A-series. It references Doc. 267, 280, 182, 279, 275 and Doc. 190 (P18–P20, P33, P35, P44, K6).

[POSIT] Thesis: Dark energy is a reading artefact — computationally present, without reading-independent consequence. Dark matter is *not* — it has reading-independent evidence and geometric candidates (P44 A–C computed, μ derivation from T^4 open).

Conventions

Legend. Every substantive claim carries a layer label in square brackets:

- **[K]** *Core derivation* — derived mathematically or numerically from the foundational posits; internally consistent and reproducible.
- **[S]** *Structural claim* — algebraically or geometrically motivated, but not yet fully proved.
- **[POSIT]** *Posit* — declared assumption or definition that is not further derived; must be accepted as such or justified separately.

P-numbers and K-numbers come from the FFGFT correction register Doc. 190 (project register). **P**n denotes an open programme point (a gap still to be closed or an ongoing computation); **K**n denotes a completed correction or anchored finding. Relevant entries in this document:

P44	MOND/rotation curves/Bullet Cluster: three-stage FE programme A/B/C for a geometric explanation of rotation curves and gravitational lensing without particle DM.
P33	Magnitude of $\xi = 4/30000$ (5^4 scale): declared intuitively/empirically motivated, no forward derivation.
P35	Bulk exponent k^*: $O(1)$ prefactor still open.
K6	H_0 as ξ^{10} power: anchored, a_0 coincidence derived.
P18–P20	Earlier open points in the cosmological sector (degeneracy, redshift, static reading).

Notation.

ξ	Dimensionless fundamental parameter, $\xi = 4/30000$; geometric consequence of the fractal Hausdorff dimension $D_f = 3 - \xi$ (posit, P33).
$\tilde{T} \cdot m = 1$	Foundational relation of FFGFT: intrinsic time $\tilde{T}$ and mass m are inversely coupled — compactification relation on the torus T^4.
T^4	Compact four-dimensional torus; carrier of the FFGFT geometry.
Λ	Cosmological constant (ΛCDM); analysed here as a reading artefact.
a_0	Characteristic acceleration scale of the MOND regime, empirically $\approx 1.2 \times 10^{-10}$ m/s^2; in FFGFT: $a_0 = cH_0/(2\pi)$ from the ξ^{10} chain.
H_0	Hubble constant; from FFGFT: $H_0 = (\pi/2)\xi^{10}\, c/\bar{\lambda}_e = 66.82$ km/s/Mpc.
$\bar{\lambda}_e$	Reduced Compton wavelength of the electron, $\bar{\lambda}_e = \hbar/(m_e c)$.
$\mu(x)$	Interpolation function of the MOND equation $a_{\text{tot}} \cdot \mu(a_{\text{tot}}/a_0) = a_{\text{Newton}}$; here: Simple form $\mu(x) = x/\sqrt{1+x^2}$ (limits from FFGFT, form a posit).
γ_{PPN}	Parametrised post-Newtonian parameter; K3 yields $\gamma_{\text{PPN}} = 1$ exactly (Cassini: $\lvert\gamma - 1\rvert < 2.3 \times 10^{-5}$).
K_{frak}	Bridge factor $K_{\text{frak}} = 1 - 100\xi = 74/75$; absorbs fractal corrections in SI absolute values.
ξ chain	Shorthand for derivation chains of the form $H_0 \propto \xi^{10}$, $a_0 \propto \xi^{10}$ etc. — all from the same ξ, without fits.
[K3]	Area-invariance rule: time-space 2-cell invariant, $L_t \cdot l =$ const; sole survivor of the T^4 selection computation (Stage B).

Chapter A

Definition: Reading Artefact

[POSIT] A quantity is called a *reading artefact* here when it simultaneously satisfies three conditions:

1. It is determined exclusively from data that are degenerate between the competing readings (Doc. 267).
2. It possesses no independent anchor outside this dataset — no laboratory measurement, no local dynamics, no obligation in a foreign sector.
3. When the reading changes, it does not disappear as *refuted* but as *objectless*: there is nothing it referred to.

Historical parallel: epicycles. They computed correctly, within their reading, and yet referred to nothing. The transition to Kepler did not falsify them but rendered them objectless. A reading artefact is not a computational error — it is a quantity without referent.

Chapter B

Dark Energy: All Three Conditions Satisfied

[K] Condition 1. All evidence for Λ is cosmological: SNe Ia luminosities, BAO angular scales, CMB acoustic peaks — exactly the three pillars whose degeneracy between the expansion and static readings Doc. 267 demonstrates in detail. No source of evidence for Λ exists outside this dataset.

[K] Condition 2. Λ has no independent anchor — and more sharply: the only theoretical candidate has failed. The QFT vacuum energy density misses the fit value by $\sim 10^{122}$ (the fine-tuning problem; see Doc. 279, outlook). ΛCDM thus introduces a quantity that (a) is not reading-independently forced by the degenerate data and (b) is missed by its own underlying theory by 122 orders of magnitude. Λ carries no obligation in any other sector of physics: it works nowhere except its own fit.

[K] Condition 3. In the static reading (Doc. 280) there is no expansion to be accelerated. The apparent acceleration is part of the degeneracy structure (Doc. 267, § accelerated dilatation in both models). When the reading changes, Λ is not refuted — it loses its object.

[K] Conclusion: Dark energy is a reading artefact in the sense of Ch. 1. Formulation with care: *not* "there is no dark energy" (that would be a statement within the expansion reading), but: "dark energy is a reading-internal bookkeeping quantity without reading-independent existence evidence."

Chapter C

Dark Matter: Condition 1 Violated — Not an Artefact

[K] Dark matter does *not* satisfy Condition 1. Its evidence lies outside the redshift degeneracy:

- **Rotation curves:** local dynamics of individual galaxies — independent of the z reading.
- **Gravitational lensing:** geometry of light propagation at individual masses — independent of the z reading.
- **Bullet Cluster:** spatial separation of lensing mass and X-ray gas in a single object — independent of the z reading.

These findings do not disappear when the reading changes. They require an explanation in *every* reading.

[POSIT] For FFGFT, dark matter is therefore not an artefact but an **open geometric problem**: the candidate is fractal path elongation (Doc. 182) as a source of additional effective attraction without a particle substrate — status: reduction layer, no core derivation. P44 Stages A–C are computed: a_0 anchor (Stage A), factor-2 gate and selection computation (Stage B), Bullet Cluster and rotation curves (Stage C). Remaining open: forward proof of K3 and form of $\mu(x)$ from T^4 geometry (P44 in Doc. 190).

Caution regarding one's own framework: The formulation "the dark sector is invented" would be sweeping and false; it would fail at the Bullet Cluster. The defensible claim distinguishes: dark *energy* — reading artefact; dark *matter* — genuine explanatory burden, geometric candidates computed, forward proof open.

Chapter D

Symmetry Check: Does the Argument Cut FFGFT Too?

[K] Doc. 267 requires symmetric application. The objection would be natural: "FFGFT's fractal time unfolding is just as computationally present and without independent consequence as Λ." Checking against the conditions of Ch. 1:

- Condition 2 is *violated* for ξ (and the artefact status is therefore excluded): ξ is cross-anchored. The same ratio carries the lepton mass ladder, the Koide scalar, the α reference, K_{frak}, and the bulk exponent ($k^* = \ln \varphi/\xi$, Doc. 275) — and only *thereafter* H_0 as ξ^{10} power (Doc. 279). A quantity with obligations in five foreign sectors cannot become objectless under a cosmological reading change; it continues to work elsewhere.
- Λ has exactly zero such external obligations.

This is the decisive asymmetry, and it is bookkeeping, not a value judgement: not "our quantity is true" but "our quantity has external obligations, Λ has none."

[POSIT] Honesty note: The magnitude of ξ (5^4 scale) is declared intuitively/empirically motivated, not a forward derivation (P33, Doc. 190). The cross-anchoring stands independently — it concerns the *consequences* of ξ, not its origin.

Chapter E

Demarcation: What This Document Does Not Claim

- It does not claim that the static reading is proved. Doc. 267 stands: no reading is currently empirically preferred.
- It does not claim that ΛCDM computes incorrectly. Reading artefacts compute correctly — that is part of the definition.
- It does not claim that dark matter is fully explained. Stages A–C are computed and passed at order-of-magnitude level. Bullet: peak 10 kpc from galaxies (mass-contrast effect). Rotation curves: DDO 154 ratio 0.998; NGC 3198/MW ratio $\sim$0.80. Open posit: interpolation form $\mu(x)$ from T^4 geometry not yet derived (P44-B).
- It provides no killer test. The contribution is ontological sorting: which quantities survive a reading change, which do not — and what that says about their status.

Chapter F

P44 as Programme: Stage Structure and Results

[POSIT] P44 is not a single computation but a three-stage programme. All three stages have been computed; their status is kept separate. Five verification scripts, all inputs declared, no fits.

F.1 Stage A — The a_0 Anchor

[K] The empirical acceleration scale below which rotation curves flatten is $a_0 \approx 1.2{\times}10^{-10}$ m/s^2 (RAR fit, McGaugh et al. 2016 — declared external reference). From the FFGFT chain Doc. 279, $H_0/c = (\pi/2)\,\xi^{10}/\bar{\lambda}_e$, without any further input:

$$H_0 = 66.82 \text{ km/s/Mpc}, \qquad \frac{cH_0}{2\pi} = 1.033 \times 10^{-10} \text{ m/s}^2, \qquad \frac{cH_0/2\pi}{a_0^{\text{emp}}} = 0.86.$$

The well-known *MOND coincidence* $a_0 \sim cH_0$ is thus in FFGFT not a coincidence but a consequence of the ξ^{10} chain (K6, Doc. 190): two independent determination paths (rotation curve fit; lepton sector via $\bar{\lambda}_e$ and ξ) meet to within 14%. Structure analogous to the pion anchor (A155/A270).

[K] Rotation curves fully computed (`ffgft_307_p36_stufeC2_rotationskurven.py`): Correct MOND equation $a_{\text{tot}} \cdot \mu(a_{\text{tot}}/a_0) = a_{\text{Newton}}$ with $\mu(x) = x/\sqrt{1+x^2}$, a_0 from ξ^{10} chain (no free parameter). Results:

Galaxy	v_{Newton}	v_{FFGFT}	v_{obs} / ratio
DDO 154 (gas-dom.)	18.5 km/s	46.9 km/s	47.0 km/s / **0.998**
NGC 3198	50.4 km/s	120 km/s	150 km/s / 0.80
Milky Way	122 km/s	175 km/s	220 km/s / 0.80

DDO 154 (cleanest test, M_b best determined): ratio 0.998. NGC 3198 and MW: ratio ~0.80, dominated by M_b uncertainty (~30%).

[K] Status of the interpolation form μ: The following is derived from FFGFT: (i) $a_0 = cH_0/(2\pi)$ as transition scale (ξ^{10} chain, Stage A); (ii) both limits: $\mu \to 1$ for $a \gg a_0$ (Newton) and $\mu \to x$ for $a \ll a_0$ (MOND limit $v^4 = GM_b a_0$). The following is *not* derived: the form of μ between the limits. The choice $\mu(x) = x/\sqrt{1+x^2}$ (Simple) is the most natural analytically-monotone interpolation with the correct limits; many forms (Standard, Simple, etc.) are empirically indistinguishable without precise profile measurements.

[POSIT] Two honesty notes. First: the form of $\mu(x)$ between the limits is a posit — the limits themselves are justified, the interpolation is not (P44-B open). Second: the exactly computed chain yields 66.82 km/s/Mpc; Doc. 280 gives 66.2 — discrepancy 0.9%, cause to be clarified.

F.2 Stage B — FE Coupling and Selection Computation

[K] The mechanism of path elongation is implemented in Doc. 026 as a finite-element model with ray tracing — it is *path-dependent by construction* (the ray accumulates delay element by element along its trajectory). The prescription for how local mass deforms the element geometry is *not* an open posit — it is the duality itself. $\tilde{T} \cdot m = 1$ means: local mass sets the local rate of each element; the intrinsic time field $T(x)$ is the deformation prescription of the mesh. Sign automatically correct: element near mass → $\tilde{T}$ smaller → slower rate → ray delayed (Shapiro anchor satisfied by construction).

[K] The Factor-2 Gate — computed (`ffgft_307_p36_stufeB1_faktor2_fe.py`): FE ray tracing through a radial element grid (30001 log nodes, linear interpolation, RK4 ray equation). Three configurations of element deformation $n = 1 + \kappa GM/(rc^2)$:

Configuration	α at solar limb	Ratio to $1.75''$
A: time rate only (duality)	$0.8750''$	0.500
B: edge stretching only (path el.)	$0.8750''$	0.500
C: rate + stretching	$1.7500''$	1.000

Result: Duality *alone* fails — it yields the 1911 half result. Fractal path elongation *alone* yields exactly half too. Together they yield exactly the measured value.

[S] Two-Halves Synthesis (candidate): The two mechanisms of the corpus are the two halves of the Einstein deflection — $\tilde{T} \cdot m = 1$ the time half (1911), fractal path elongation the space half (1911 → 1915). They do not replace each other; they are complementary.

[K] Selection computation under T^4 invariance rules (`ffgft_307_p36_stufeB2_kappa_herleitung.py`): Starting point $m(r) = m_0(1 + \Phi/c^2)$ (dual reading of the measured gravitational time dilation, Pound–Rebka anchored; mass decreases in the potential well), rate $\tilde{T} = 1/m$, so $\kappa_{\text{time}} = 1$ fixed. Four candidates for the spatial edge rule, all tested in the same FE setup:

Rule	γ	α	Verdict
K1 conformal ($l = \lambda_C$, same source)	-1	$0.0000''$	excl.
K2 unimodular (4-vol. invariant)	$+1/3$	$1.1667''$	excl.
K3 area (time-space 2-cell invariant)	$+1$	$1.7500''$	**passes**
K0 rate only (1911 reference)	0	$0.8750''$	excl.

The conformal trap: The most natural path — edge length equal to Compton length $\hbar/mc$ of the *same* node — makes the geometry conformally flat: light sees nothing, $\alpha = 0$. Harder excluded than the half result.

[S] The survivor and its form: K3 reads $L_t \cdot l = \text{const}$ — area invariance of the time-space 2-cell. This is formally the reciprocity form of $\tilde{T} \cdot m = 1$, applied a second time, now to the time-space pair; the corpus formula "c = space-time bridge" thereby receives a precise reading: c as the invariant area of the cell. Consequence: $\gamma_{\text{PPN}} = 1$ *exactly* (Cassini passed with unlimited margin; K1/K2 lie a factor 10^5 resp. 3×10^4 above the bound). Shapiro consistency numerically validated (rel. dev. 8×10^{-5} against the analytic path-integral form).

[POSIT] Status and limits: This is an exclusion proof over a *declared* candidate list plus one exactly passing survivor — not a forward proof. Remaining open: why the 2-cell area (and not the 4-volume) is the invariant of the T^4 metric. Completeness of the candidate list is a posit.

[POSIT] Safeguards against self-deception: (i) The characteristic scale $r_\xi(M)$ must *not* be set to the Compton wavelength of the total mass ($\hbar/Mc \sim 10^{-68}$ m for a galaxy) — with rescue exponents that would be free-parameter renaming; the scale must follow from the Doc. 182 geometry. (ii) Expressions of the form $\Phi_{\text{frak}} \propto \int (d\xi/dr)\, dr$ are empty identities without derivational content and will not be admitted to the corpus. Profile candidates without derivation are, if used, to be declared as posits with their own parameter.

F.3 Stage C — Bullet Cluster and Rotation Curves: Computed

[K] Bullet Cluster requirement from published lensing maps (Clowe et al. 2006, ApJ 648 L109): lensing mass at the subcluster galaxy position $\sim 2.3 \times 10^{14}\, M_\odot$, baryons there ~10%, spatial offset lensing peak – gas peak ~250 kpc.

[POSIT] For Doc. 182 the expectation was: the ξ-field profile would follow the gas density at instantaneous coupling — and thus fail at the Bullet. The computation shows the opposite.

[K] Sharpening: Pre-collision, the baryonic potential of both clusters is *gas-dominated* (~85% of baryons are ICM gas; stars/galaxies ~15%). A pure potential-*memory* field would remember gas-dominated potential wells — it would smear at the collision site, not peak at the *current* galaxy position. Memory alone is insufficient: the field would need to be *advected* with the collision-free component.

[K] Test setup and result (`ffgft_307_p36_stufeC1_bullet_cluster.py`): Two-clump mesh, projection along line of sight (weak lensing), $\tilde{T} \cdot m = 1$ (K3, instantaneous coupling). Parameters directly from Clowe+2006: $M_{\text{gal}} = 2.3 \times 10^{14}\, M_\odot$, $M_{\text{gas}} = 0.23 \times 10^{14}\, M_\odot$, offset 250 kpc.

Quantity	Value	Requirement
Lensing peak position (FFGFT)	$x = 10.0$ kpc	at galaxies
Distance peak – galaxies	10 kpc	< 50 kpc
Distance peak – gas	240 kpc	> 150 kpc

[K] Verdict: Bullet test passed — unexpected at instantaneous coupling. **Cause:** $M_{\text{gal}}/M_{\text{gas}} = 10$ (observed). The galaxy potential dominates the gas field despite a 250 kpc offset. Sensitivity test with $M_{\text{gal}} = M_{\text{gas}}$ (pessimistic): peak at 30 kpc — still at the galaxies. The profile radius decides, not the advection history.

[POSIT] Honesty note: This is a 1D projection test, not a full 2D lensing reconstruction. The decisive factor is the mass ratio 10, not the mechanism itself. A sharper test requires: (i) observed 2D lensing profile against computation; (ii) test with sub-clusters where $M_{\text{gal}}/M_{\text{gas}}$ is smaller. The collapse risk of field-DM is absent at instantaneous coupling: the field follows the instantaneous density, no advected field needed.

Chapter G

Layer-Status Summary

- **[K]** Degeneracy of the three data pillars: taken from Doc. 267.
- **[K]** Λ satisfies all three artefact conditions (Ch. 2).
- **[K]** DM violates Condition 1; evidence reading-independent (Ch. 3).
- **[K]** Asymmetry ξ vs. Λ via cross-anchoring (Ch. 4).
- **[K]** Stage A: a_0 anchor $cH_0/2\pi = 1.033 \times 10^{-10}$ m/s^2, 14% deviation from RAR value (Ch. 6, §1).
- **[K]** Stage B: factor-2 gate (each $0.875''$ time and space, together $1.750''$); K3 sole survivor of the selection computation, $\gamma_{\text{PPN}} = 1$ exactly (Ch. 6, §2).
- **[K]** Stage C: Bullet test passed (peak 10 kpc from galaxies, mass-contrast effect). Rotation curves: DDO 154 ratio 0.998; NGC 3198/MW ratio ~0.80 (M_b uncertainty dominates). μ limits justified, interpolation form posit (Ch. 6, §3).
- **[POSIT]** Artefact definition (Ch. 1); DM as open geometric problem with candidate Doc. 182 (Ch. 3); P44 programme A/B/C fully computed; μ limits justified, interpolation form and K3 forward proof open (Ch. 6).

Chapter H

Open Points (Doc. 190 P44)

1. **P44-B remaining:** Factor-2 gate computed (time+space each $0.875''$, together $1.750''$); selection computation completed (K3 area invariance as sole survivor, $\gamma = 1$ exactly). Remaining: forward proof that the time-space 2-cell area (and not the 4-volume) is the invariant of the T^4 metric — plus completeness check of the candidate list.
2. **P44-C Bullet:** Computed (`ffgft_307_p36_stufeC1_bullet_cluster.py`). Contrary to expectation, passed: $M_{\text{gal}}/M_{\text{gas}} = 10$ dominates at instantaneous coupling. Lensing peak 10 kpc from galaxies, 240 kpc from gas. Collapse risk field-DM absent (instantaneous coupling without advection). Open refinement: 2D lensing reconstruction, sub-clusters with smaller mass contrast.
3. **P44-C Rotation curves:** Computed (`ffgft_307_p36_stufeC2_rotationskurven.py`). DDO 154: ratio 0.998 (passed). NGC 3198/MW: ratio ~0.80 (M_b uncertainty dominates, not mechanism). Open posit: form of $\mu(x)$ between the limits not derived from T^4 geometry (limits themselves justified, interpolation open — P44-B).
4. H_0 **discrepancy:** 66.82 vs. 66.2 km/s/Mpc — cause to be clarified in Doc. 279/280 (rounding/m_e version).
5. Relation of the artefact analysis to the early universe phase (BBN, structure formation, Doc. 267 § honest position) — there, one reading might yet acquire reading-independent burdens.
6. Check whether the artefact definition (Ch. 1) applies to further ΛCDM quantities (inflaton parameters?) — and symmetrically to FFGFT quantities in the reduction layer.

Chapter I

References and External Sources

Internal documents: Doc. 267 (cosmological degeneracy); Doc. 280 (achromatic static redshift, K6); Doc. 182 (fractal path elongation, Doc. 026 FE implementation); Doc. 279 (H_0 as ξ power, Λ fine-tuning in outlook); Doc. 275 (k^* recursion); Doc. 190 (P18–P20, P33, P35, P44, K6).

Verification scripts: `ffgft_307_p36_stufeA_a0_anker.py` (all inputs declared, no fits); `ffgft_307_p36_stufeB1_faktor2_fe.py` (30001 log nodes, RK4, convergence checked); `ffgft_307_p36_stufeB2_kappa_herleitung.py` (selection computation, four candidates); `ffgft_307_p36_stufeC1_bullet_cluster.py` (two-clump projection, Clowe+2006 parameters); `ffgft_307_p36_stufeC2_rotationskurven.py` (DDO 154, NGC 3198, MW, no free parameters).

Declared external references: McGaugh, Lelli, Schombert 2016, PRL 117, 201101 (RAR/a_0); Clowe et al. 2006, ApJ 648, L109 (Bullet Cluster).